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AI Developer

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Hiring Team
CT19
Germany (remote)

Essen, November 13, 2026

Application for Machine Learning Data Scientist - BioML

Dear Mason and the CT19 team,

I am writing to apply for the Machine Learning Data Scientist - BioML position at CT19. I have a Master's in Artificial Intelligence and three years of experience building and deploying AI applications, including models that predict process outcomes from sensor time series in chemical production.

At Covestro I worked as a Data Scientist (Intern) on predicting the outcome of prepolymer synthesis, such as viscosity and reaction progression, from time-series process data. I compared Random Forest, XGBoost and an LSTM, built lag features and rolling windows from the raw sensor signals, and benchmarked everything against the empirical curve-fitting the chemists already used. The models flagged batches likely to miss specification earlier, and a Streamlit dashboard let chemists try parameter changes and see the predicted result before running a physical batch. Earlier, at Cognizant, I built ARIMA, SARIMA, XGBoost, Prophet and LSTM models for EV load forecasting.

Mechanistic modelling is where I have the most to learn: I have not built kinetic ODE or genome-scale metabolic models in a job, and I have not worked with Raman or NIR spectra. Time-series modelling of process data, evaluation against ground truth and shipping the result as a tool are the parts I can start contributing to right away.

My recent work is on making model output trustworthy for domain experts. At Kniit I am the only developer on a knowledge-extraction pipeline, evaluated against ground truth from SAP systems, where precision and recall went from 33% and 45% to 93% and 90%. At Covestro I automated the evaluation of a RAG system against chemist-written answers. I have a co-authored conference paper, and I am used to a start-up setting where I own the work from data to a deployed tool.

CT19's idea of using lab-generated data together with predictive models to cut costly experiments and reach manufacturing faster is a problem where better models save real time and money, and the role gives a say in research direction. I am based in Germany, comfortable working remotely, with an EU Blue Card and no sponsorship needed, available from 15 November 2026, and happy to travel within Europe.

Thank you for considering my application.

Kind regards,

Chandu Prasad Bhairapu